Extensive form games and backwards induction

Practice problems

The answers are in the second half of this document.

1. In George and Noel's break up game, Noel's preferences were formulated with the idea that (1) she prefers breaking up to not breaking up and (2) if they are not going to break up, she'd prefer not to have George try to break up with her. But, later, when George does try to break up with her, she resists it.[1] So, the solution that we found was not the actual outcome.

Keep the tree as it is in figure 1 in "The preemptive break up and extensive form games" and keep George's payoffs the same, but change Noel's so that (1) if George doesn't break up with her, she prefers breaking up to staying together, but (2) if George does try to break up with her, she prefers staying together to breaking up.

This can be done using variables instead of specific utility numbers. Now, what is the solution for this game?

2. In this game, there is the kidnapper, whom we'll call Guy, and the victim's wife, Vivica, who has been contacted to pay ransom. Although the victim (whom we'll name Orlando) is surely affected by what transpires, we are presuming that the victim has no options.

Guy's choices are to kidnap (Orlando) and do not kidnap. If he chooses the latter, then the tree comes to an end, which represents "game over." If, instead, he chooses to kidnap Orlando, then Vivica is informed of the kidnapping and decides whether to pay the ransom. In response to Vivica's decision, Guy decides whether to release Orlando or kill him.

(from Harrington, *Games, Strategies, and Decision Making*, pp. 18 - 19)

There are five possible outcomes. If we list them from most to least preferred for Vivica and assign ordinal utility values, they are as follows.

G. doesn't kidnap O.; V. doesn't have to do anything:	(don't kidnap, ---)	5
G. kidnaps and releases O.; V. doesn't pay the ransom:	((kidnap, release), don't pay)	4
G. kidnaps and releases O.; V. pays ransom:	((kidnap, release), pay)	3
G. kidnaps and kills O.; V. doesn't pay the ransom:	((kidnap, kill), don't pay)	2
G. kidnaps and kills O.; V. pays the ransom:	((kidnap, kill), pay)	1

For the kidnapper, Guy, the outcomes are listed this way:

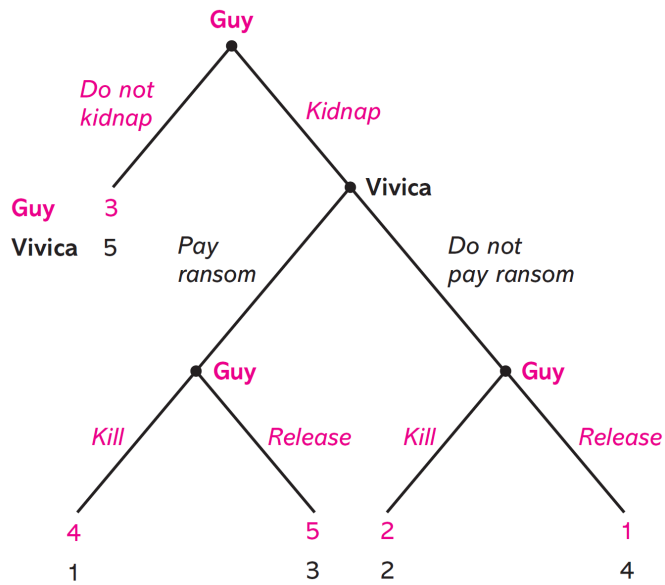
G. kidnaps and releases O.; V. pays ransom:	((kidnap, release), pay)	5
G. kidnaps and kills O.; V. pays the ransom:	((kidnap, kill), pay)	4

[1] Although, naturally, by the end of the episode, she has broken up with him.

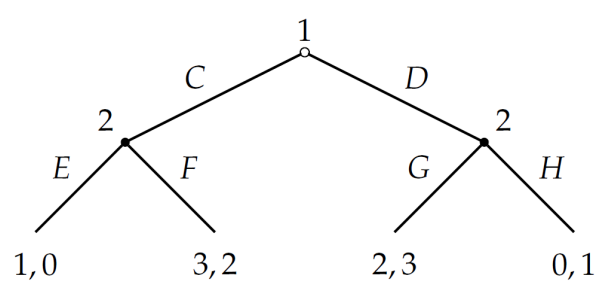
G. doesn't kidnap O.; V. doesn't have to do anything:	((don't kidnap, ---)	3
G. kidnaps and kills O.; V. doesn't pay the ransom:	((kidnap, kill), don't pay)	2
G. kidnaps and releases O.; V. doesn't pay the ransom:	((kidnap, release), don't pay)	1

If you want to try creating the tree for this game, don't look any further. Use backward induction to find the solution to the game.

Here is the tree. Guy's payoff is the top number in each pair. Vivica's is the bottom one.

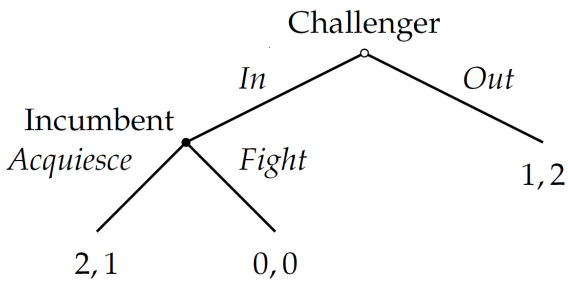


3. Use backward induction to find the solution to this game.



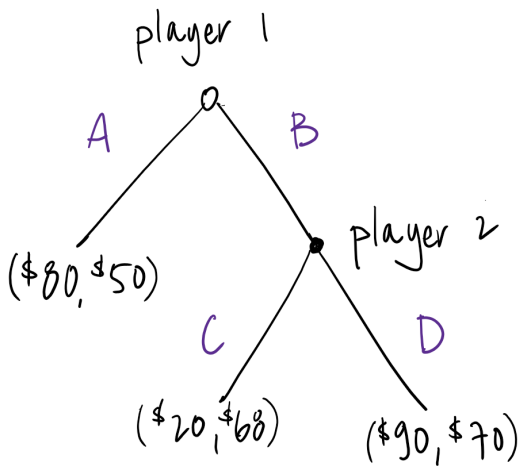
(from M. Osborne, *An introduction to game theory*)

4. Use backward induction to find the solution to this game.



(from M. Osborne, *An introduction to game theory*)

5. Use backwards induction to find the solution to this game.



Answers

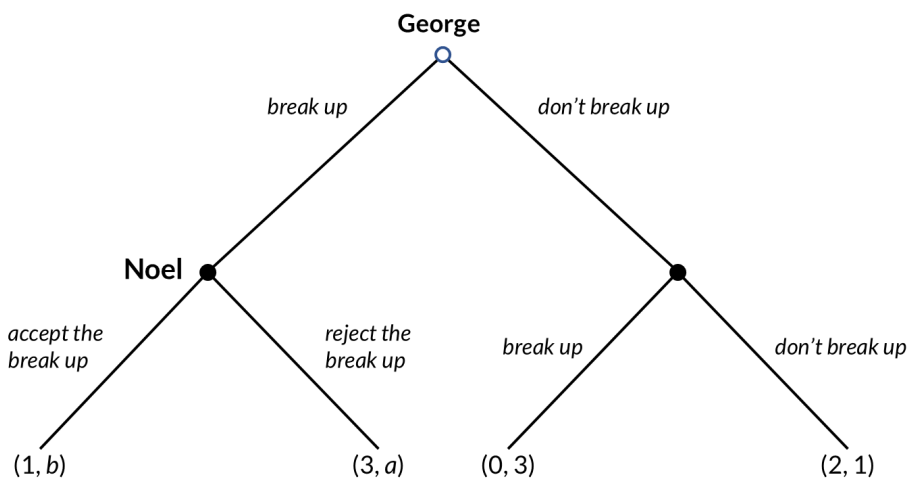
1. Now, Noel has these two preferences:

1. (G doesn't, she breaks up with him) **P** (G doesn't, she doesn't)
2. (G does, she rejects it) **P** (G does, she accepts it)

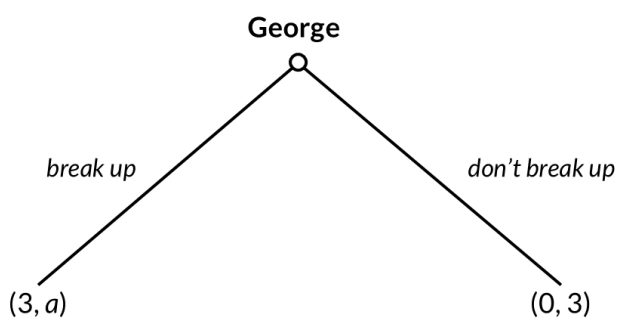
The first one was one of Noel's preferences in our original list for her, and so the values for those outcomes can stay as they were, $3 > 1$—which is all that we need to proceed.

For (2), we can use the variables a and b with $a > b$.

The game tree, then, is this:

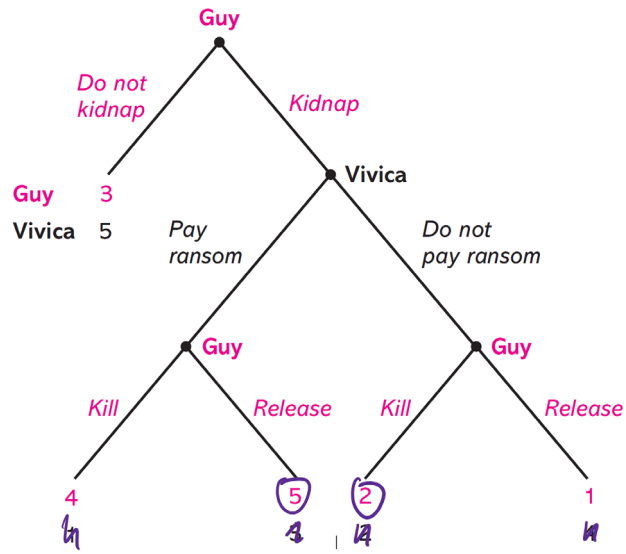


Payoff a is greater than payoff b, and 3 is greater than 1. So, Noel if George chooses *break up*, she will choose *reject it*. And if George chooses *don't break up*, she will choose *break up*. Thus, the game reduces to this:

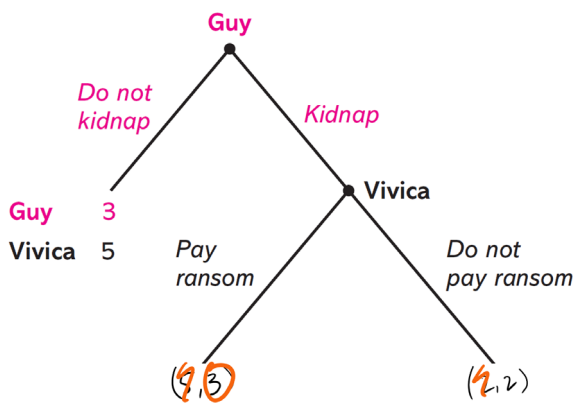


For George, 3 is greater than 0, and so he will choose *break up*. Therefore, the solution to the game and the Nash equilibrium is (*George breaks up with Noel, she rejects it*).

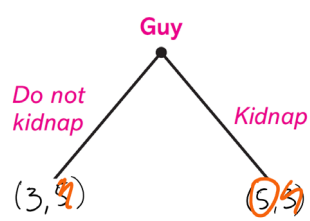
2. Starting at the bottom of the tree, if Vivica pays the ransom, Guy will release Orlando (because 5 > 4). If Vivica does not pay the ransom, Guy will kill Orlando (because 2 > 1).



So, the game reduces to this, and we find that if Guy kidnaps Orlando, Vivica will pay the ransom (because 3 > 2).



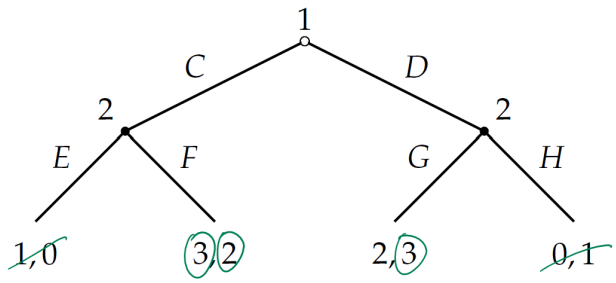
This means that the game reduces to this, and Guy will choose to kidnap Orlando (because 5 > 3).



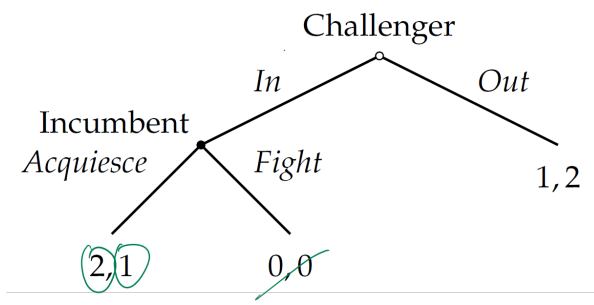
If we retrace all of that, then Guy kidnaps Orlando, Vivica pays the ransom, and Guy releases Orlando—((kidnap, release), pay) is the solution.

3. If player 2 has to choose between E and F, he or she will choose F (because 2 > 0). If player 2 has to choose between G and H, he or she will choose G (because 3 > 1). So, when player 1 chooses between C

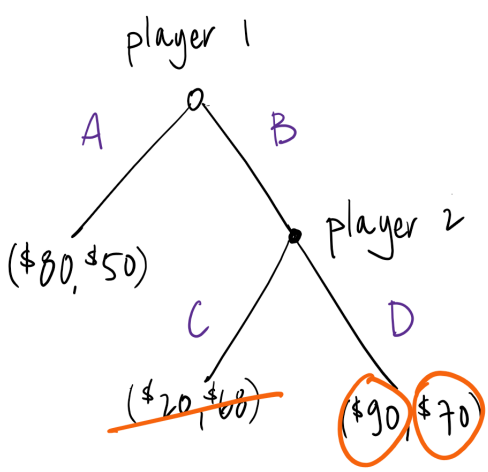
and D, the outcome for C is, effectively, (3, 2) , and the payoff for D is (2, 3). Thus, player 2 will choose C, which makes the solution (C, F).



4. If the incumbent has to choose between *acquiesce* and *fight*, he or she will choose *acquiesce* (because 1 > 0). Thus, when the challenger is deciding between *In* and *Out*, the outcome if he or she chooses *In* is (2, 1). This means that the challenger will choose *In* (because 2 > 1), which makes the solution (*In*, *Acquiesce*).



5. If player 2 has to choose between C and D, he or she will—or should—choose D (because \$70 > \$68). That makes player 1's decision a choice between getting \$80 or getting \$90, and so, he or she will choose B. Thus, the solution is (B, D).



One concern with this game, however, is that when (or if) player 2 is making a choice between C and D, he or she may accept a small penalty (that is, giving up \$2) and choose C to keep player 1 from getting \$90 (while he or she is stuck with a choice between \$68 or \$70). If player 1 suspects that this might

happen, then he or she will choose A instead of B (to get $80 instead of $20). If this happens, then player 1 ends up with $80 (and not $90) and player 2 ends up with $50 (and not $70).